

Analysis of FlexCast Reconfiguration

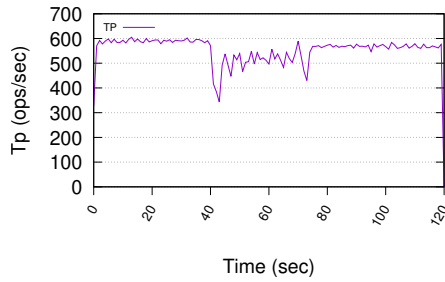
1 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 2 min
- Reconfig interval: 30sec
- Regions/Nodes: 3
(Canada, Virginia, Sao Paulo)
- Clients: 150 (50 per region)
- Locality: 95%

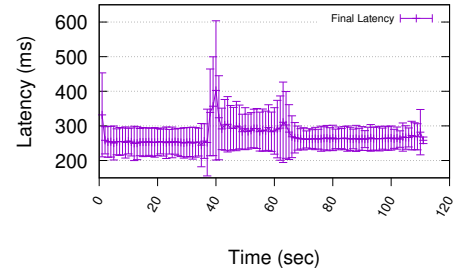


2 Results

2.1 Throughput



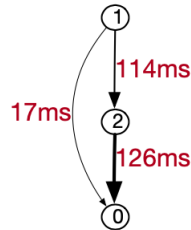
2.2 Latency



2.3 Reconfiguration details

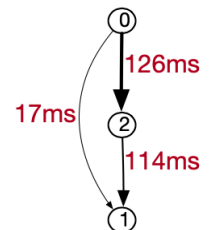
```
workload 1 : [
  {
    "destination": [0, 1],
    "percentage": 0.032794576
  },
  {
    "destination": [0, 2],
    "percentage": 0.57712483
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.050535202
  },
  {
    "destination": [1, 2],
    "percentage": 0.3395454
  },
]
```

Took 55 ms

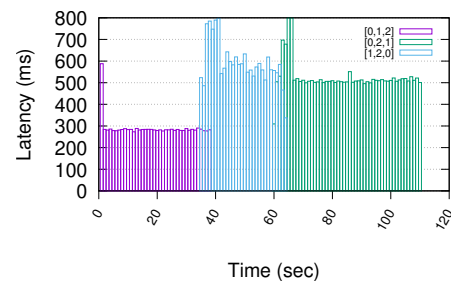
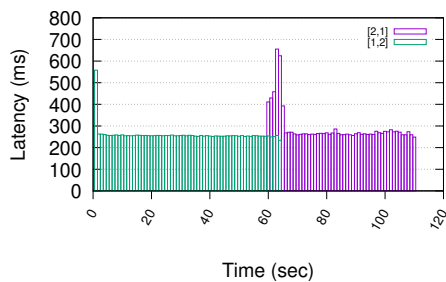
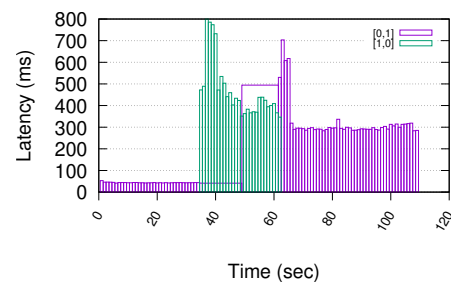
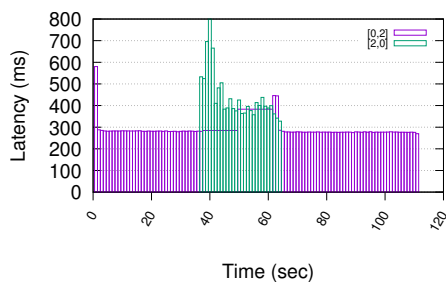


```
workload 2 : [
  {
    "destination": [1, 2, 0],
    "percentage": 0.05174473
  },
  {
    "destination": [2, 0],
    "percentage": 0.5442484
  },
  {
    "destination": [1, 0],
    "percentage": 0.034562822
  },
  {
    "destination": [1, 2],
    "percentage": 0.36944407
  },
]
```

Took 5 ms

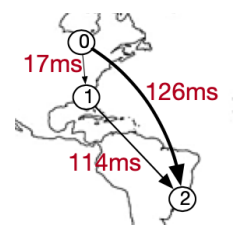


2.4 Latencies per dests (90p)



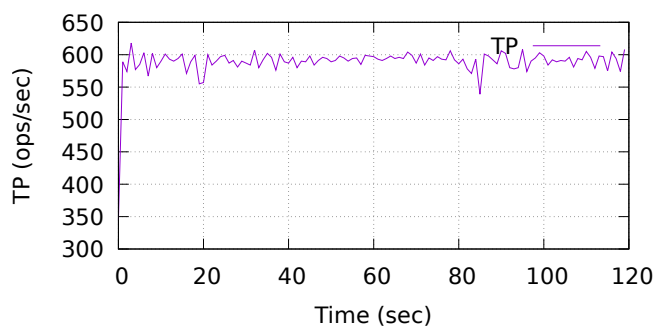
3 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 2 min
- Decided not to reconfigure
- Regions/Nodes: 3
(Canada, Virginia, Sao Paulo)
- Clients: 150 (50 per region)
- Locality: 95%

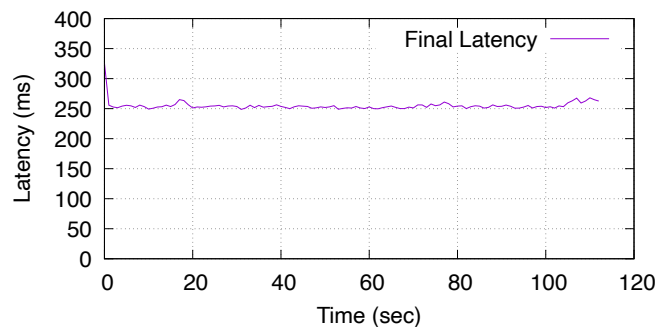


4 Results

4.1 Throughput

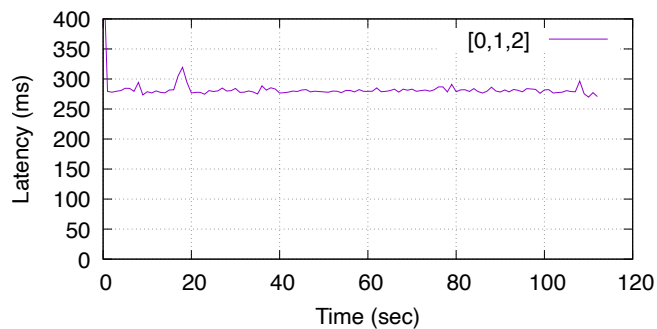
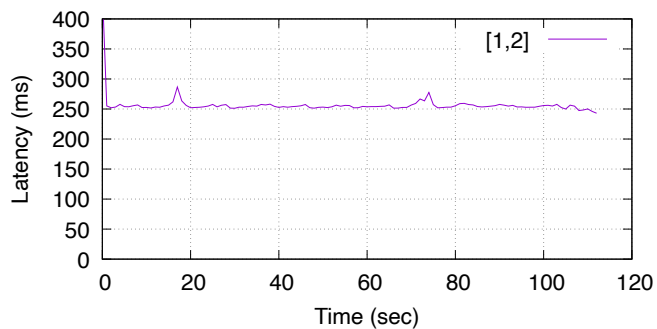
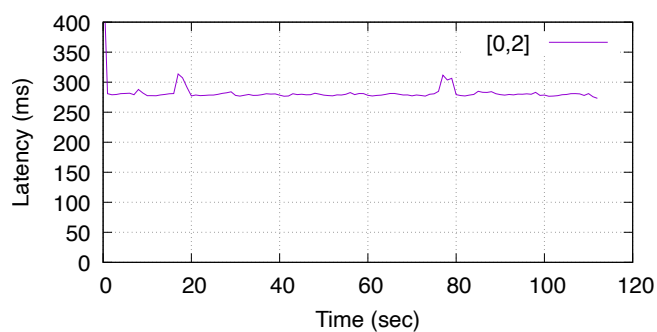
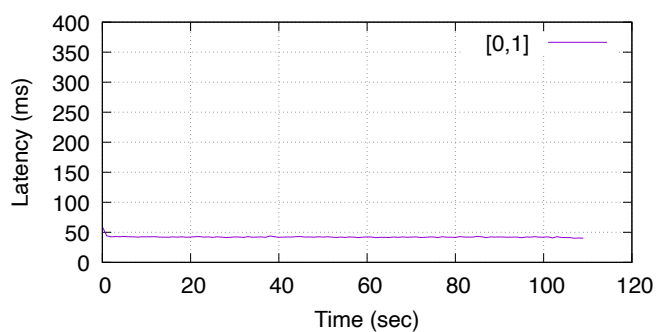


4.2 Latency



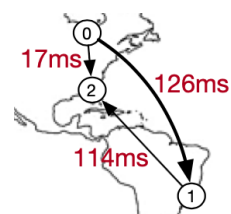
4.3 Reconfiguration details

4.4 Latencies per dests (90p)



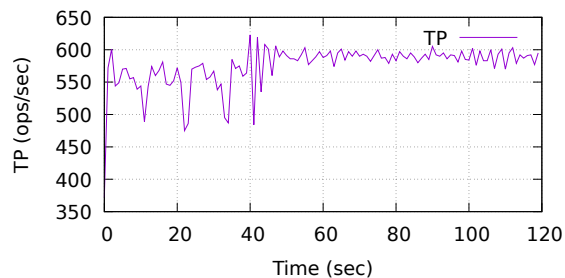
5 Experiment Config.

- **TPCC/AWS Latencies**
- **Exp. Duration:** 2 min
- **Reconfig:** 30 seconds
- **Regions/Nodes: 3**
(Canada, Sao Paulo, Virginia)
- **Clients: 150** (50 per region)
- **Locality: 95%**

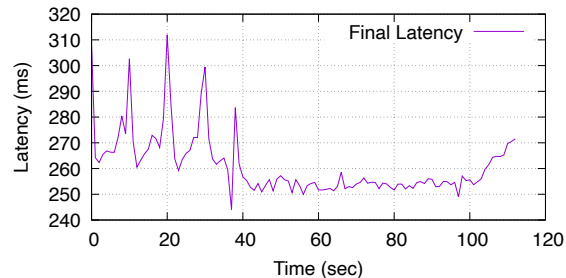


6 Results

6.1 Throughput



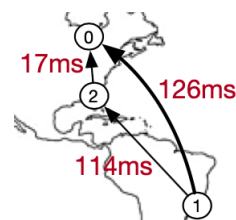
6.2 Latency



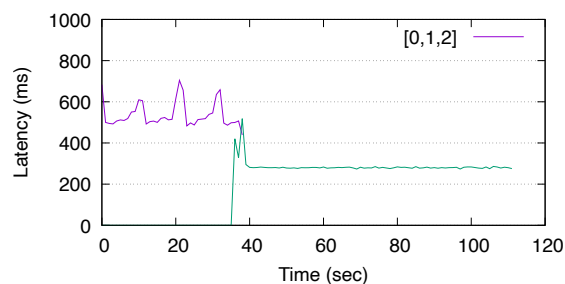
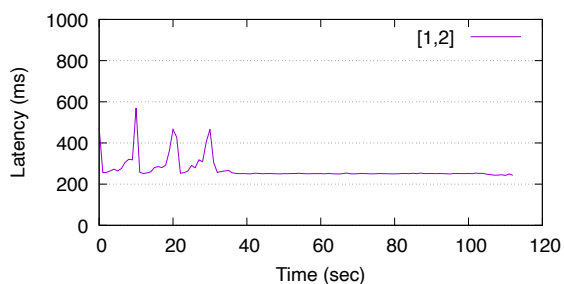
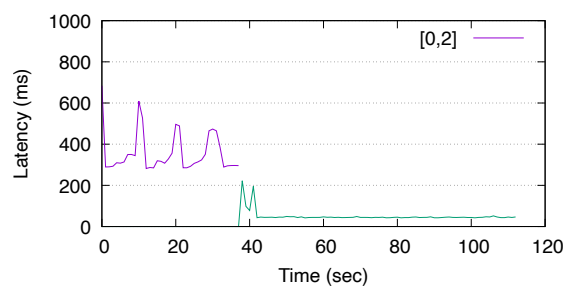
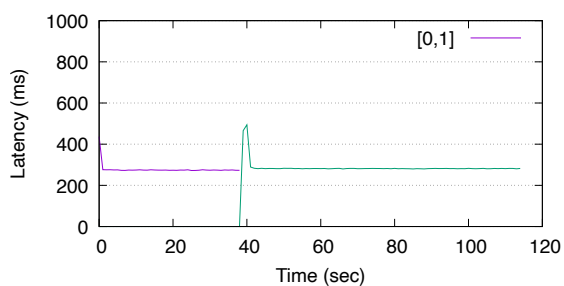
6.3 Reconfiguration details

```
"workload": [
  {
    "destination": [0, 1],
    "percentage": 0.59979254
  },
  {
    "destination": [0, 2],
    "percentage": 0.032745477
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.049208425
  },
  {
    "destination": [1, 2],
    "percentage": 0.31825358
  }
]
```

Took 24 ms

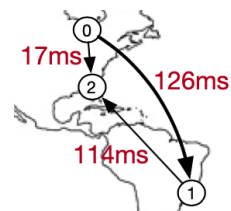


6.4 Latencies per dests (90p)



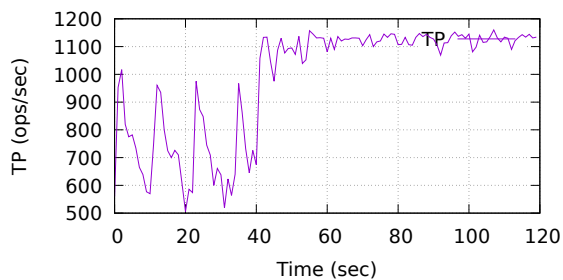
7 Experiment Config.

- TPCC/AWS Latencies
NO CLIENT LATENCIES
- Exp. Duration: 2 min
- Reconfig: 30 seconds
- **GC: 10 seconds**
- **Regions/Nodes: 3**
(Canada, Sao Paulo, Virginia)
- **Clients: 150** (50 per region)
- **Locality: 95%**

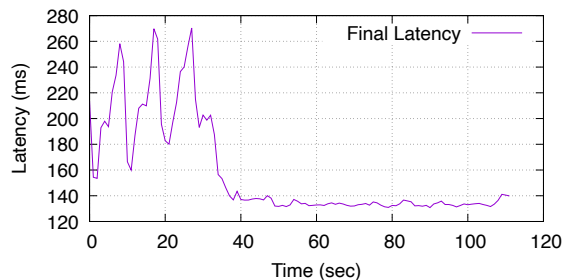


8 Results

8.1 Throughput



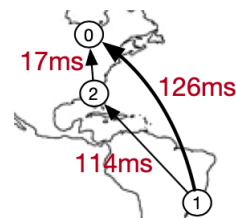
8.2 Latency



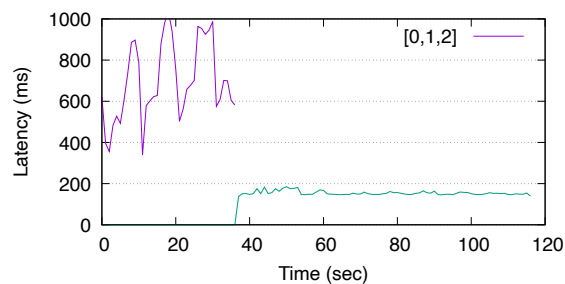
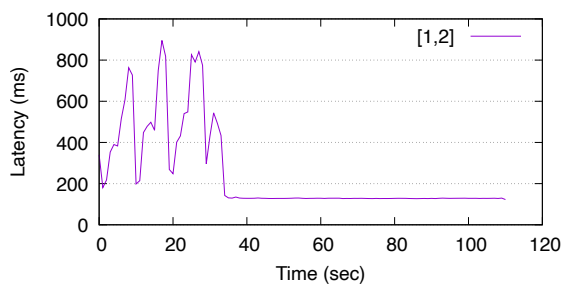
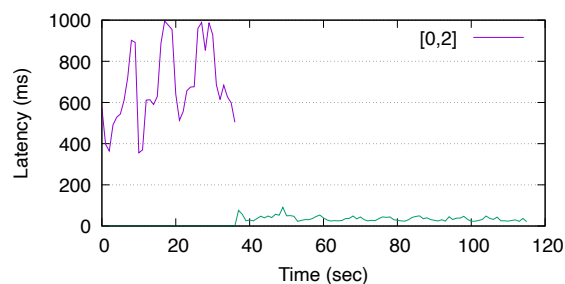
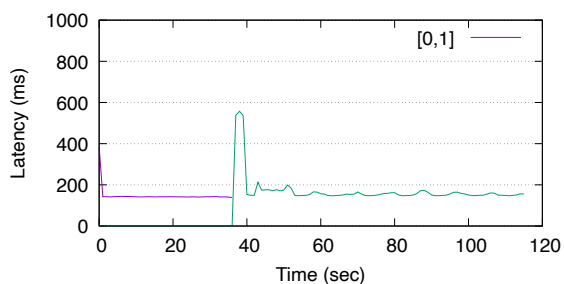
8.3 Reconfiguration details

```
"workload": [
  {
    "destination": [0, 1],
    "percentage": 0.714834
  },
  {
    "destination": [0, 2],
    "percentage": 0.029263614
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.047801804
  },
  {
    "destination": [1, 2],
    "percentage": 0.2081006
  }
]
```

Took 22 ms

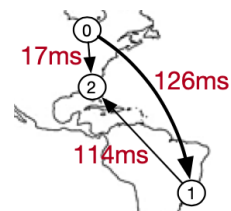


8.4 Latencies per dests (90p)



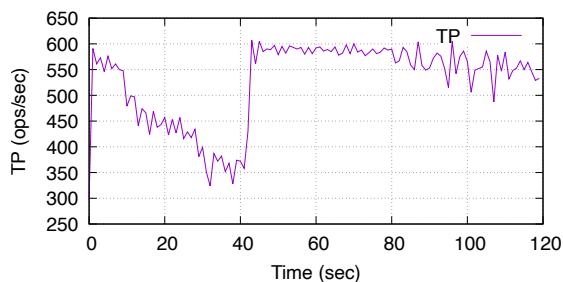
9 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 2 min
- Reconfig: 30 seconds
- **NO GC**
- **Regions/Nodes: 3**
(Canada, Sao Paulo, Virginia)
- **Clients: 150** (50 per region)
- **Locality: 95%**

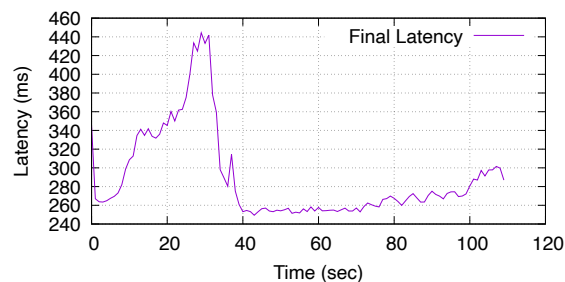


10 Results

10.1 Throughput



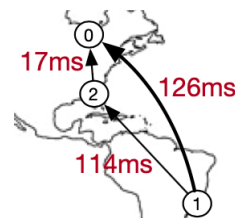
10.2 Latency



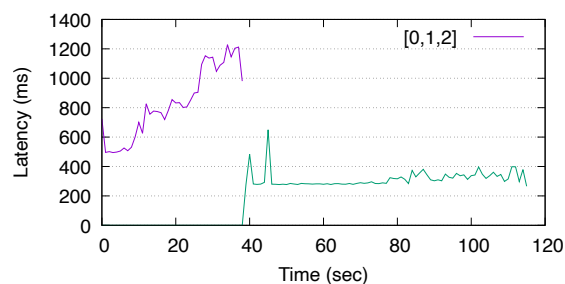
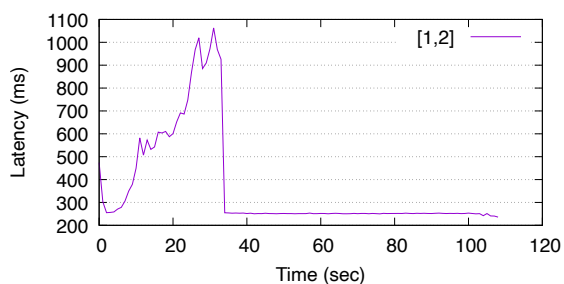
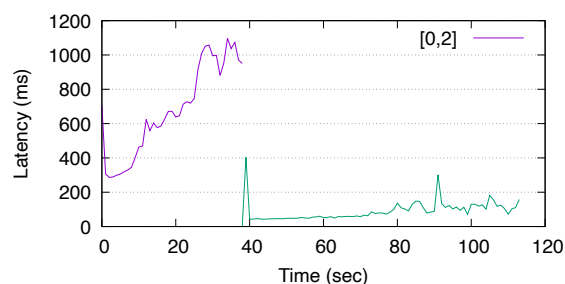
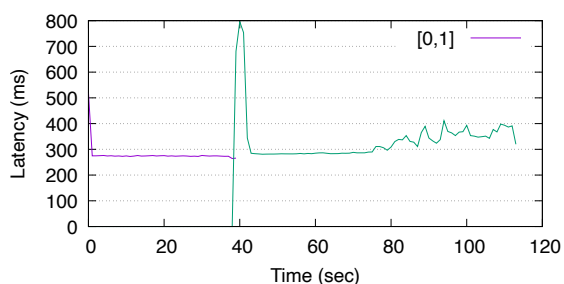
10.3 Reconfiguration details

```
"workload": [
  {
    "destination": [0, 1],
    "percentage": 0.6664093
  },
  {
    "destination": [0, 2],
    "percentage": 0.029835107
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.05035019
  },
  {
    "destination": [1, 2],
    "percentage": 0.2534054
  }
]
```

Took 25 ms

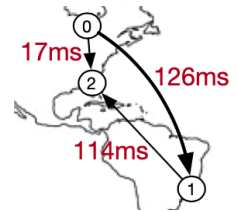


10.4 Latencies per dests (90p)



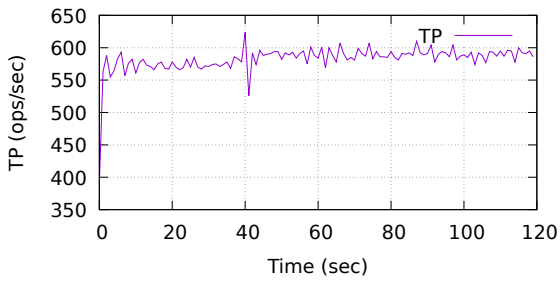
11 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 2 min
- Reconfig: 30 seconds
- GC 1 second
- Regions/Nodes: 3
(Canada, Sao Paulo, Virginia)
- Clients: 150 (50 per region)
- Locality: 95%

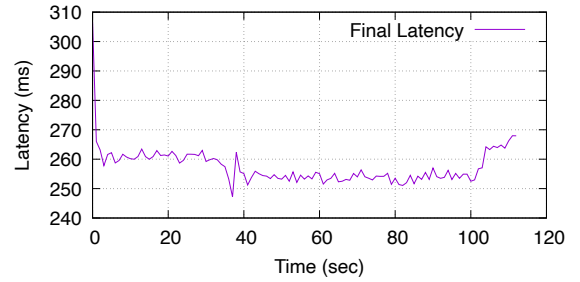


12 Results

12.1 Throughput



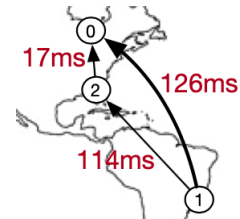
12.2 Latency



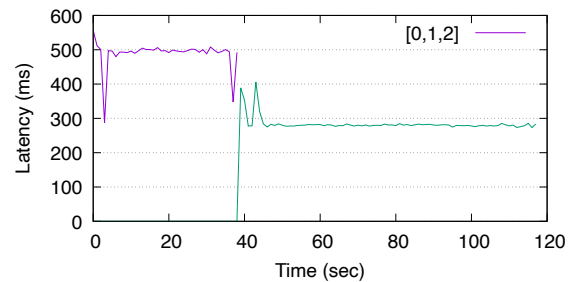
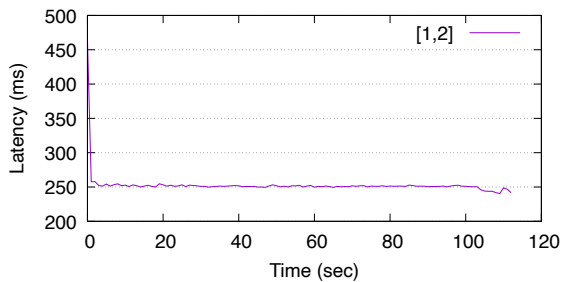
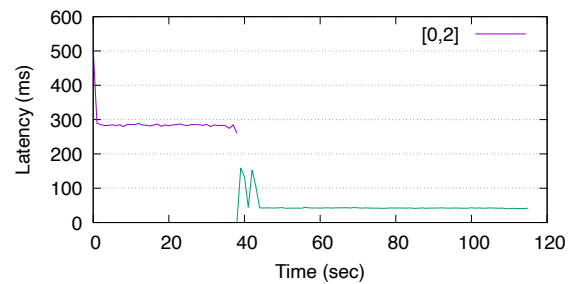
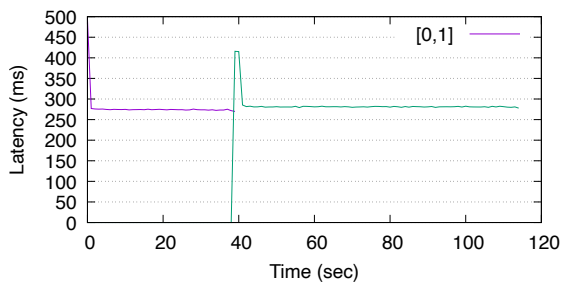
12.3 Reconfiguration details

```
"workload": [
  {
    "destination": [0, 1],
    "percentage": 0.5822697
  },
  {
    "destination": [0, 2],
    "percentage": 0.031214725
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.050273508
  },
  {
    "destination": [1, 2],
    "percentage": 0.33624208
  },
]
```

Took 22 ms

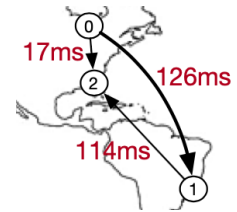


12.4 Latencies per dests (90p)



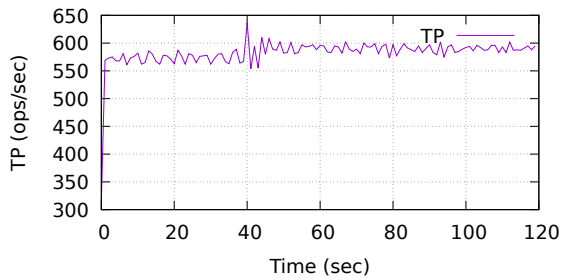
13 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 2 min
- Reconfig: 30 seconds
- GC 2 second
- Regions/Nodes: 3
(Canada, Sao Paulo, Virginia)
- Clients: 150 (50 per region)
- Locality: 95%

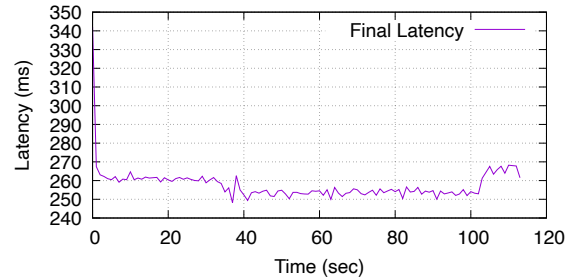


14 Results

14.1 Throughput



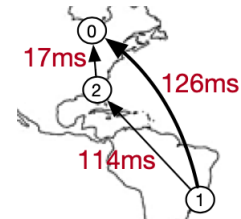
14.2 Latency



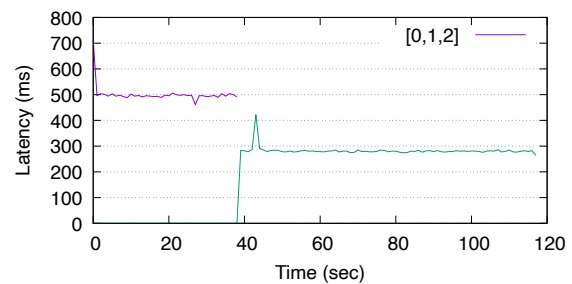
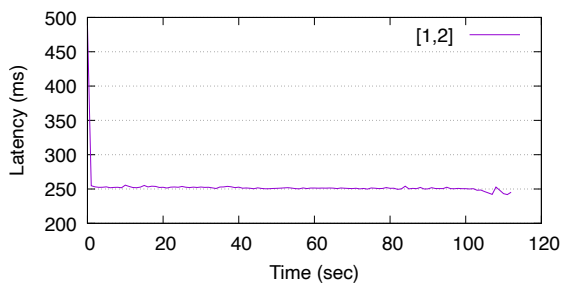
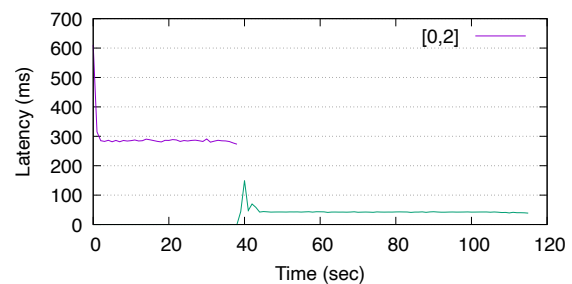
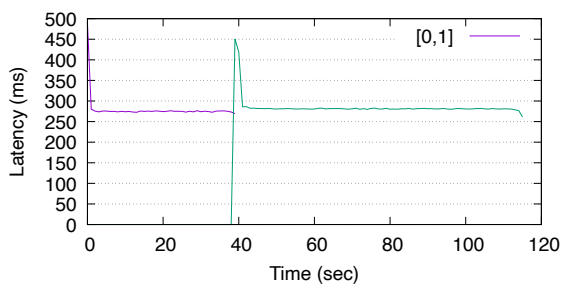
14.3 Reconfiguration details

```
"workload": [
  {
    "destination": [0, 1],
    "percentage": 0.58594364
  },
  {
    "destination": [0, 2],
    "percentage": 0.03219614
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.04894861
  },
  {
    "destination": [1, 2],
    "percentage": 0.3329116
  }
]
```

Took 22 ms

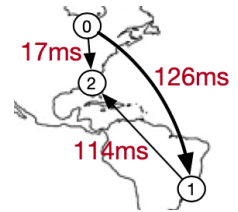


14.4 Latencies per dests (90p)



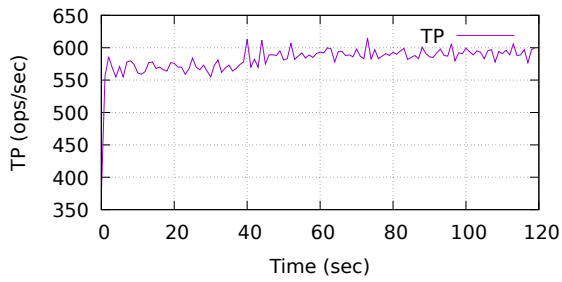
15 Experiment Config.

- TPCC/AWS Latencies
- Exp. Duration: 2 min
- Reconfig: 30 seconds
- GC 5 seconds
- Regions/Nodes: 3
(Canada, Sao Paulo, Virginia)
- Clients: 150 (50 per region)
- Locality: 95%

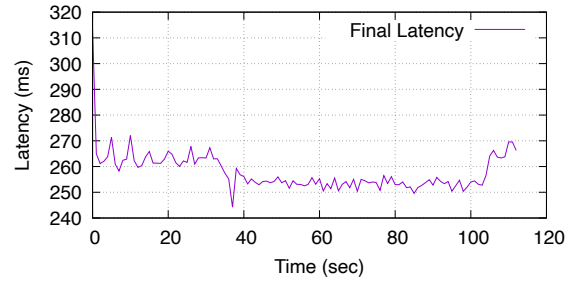


16 Results

16.1 Throughput



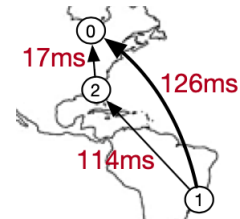
16.2 Latency



16.3 Reconfiguration details

```
"workload": [
  {
    "destination": [0, 1],
    "percentage": 0.58656037
  },
  {
    "destination": [0, 2],
    "percentage": 0.03316103
  },
  {
    "destination": [0, 1, 2],
    "percentage": 0.049413003
  },
  {
    "destination": [1, 2],
    "percentage": 0.3308656
  },
]
```

Took 22 ms



16.4 Latencies per dests (90p)

